

Victor Yang

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PERSONAL SUMMARY

I am a 4th year manufacturing engineering student at UBC with co-op experience. I have a proven track record in industry-partnered research and multidisciplinary leadership roles, I've developed strong problem-solving, leadership, and teamwork skills. I'm eager to apply both my technical and collaborative knowledge in a real-world professional engineering environment where I can continue to grow and deepen my understanding about the industry.

TECHNICAL SKILLS

Engineering / Data: Python (NumPy, SciPy, Pandas, Open3D, TensorFlow) | C++ | R | Java | IEC 61131-3
Software Packages: SolidWorks (CSWP) | Siemens NX | Microsoft Office | Gantt Project | MS Project | Git | Arena Simulation
Manufacturing: Soldering | Hand/Power Tools | Microcontrollers | 3D printing | Metrology | CNC machining | GD&T

EDUCATION

Bachelor of Applied Science (BASc) | Manufacturing Engineering
University of British Columbia

Expected graduation: April 2028

PROFESSIONAL WORK EXPERIENCE

Hanon Systems

May 2026 - Present

Engineering Co-op Student

- Facilitated a large-scale facility relocation, mapping all dependencies from early infrastructure prep to final assembly line moves, by building and maintaining a master timeline in MS Project and cost tracking dashboard in Excel.
- Prevented overall project delays and kept the relocation schedule strictly on track despite external supply chain shortages by analyzing dependencies and shifting contractor schedules to parallel workflows.
- Supported a \$100M USD cost reduction initiative by centralizing historical manufacturing data and building clear presentation decks for executive leadership.
- Ensured strict adherence to deadlines across multiple departments by facilitating weekly update meetings and managing and active open issues list to track progress from senior team members and contractors.

Composites Research Network Lab

February 2025 – May 2026

Undergraduate Research Assistant

- Led material selection for an industry partnered circular economy project, validating a recyclable, food-grade alternative with 500°F+ thermal stability by conducting thermogravimetric analysis and prototype testing.
- Automated the quantification of composite defects from 3D point cloud data by developing a custom Python program using Open3D, NumPy, and SciPy for a PHD research initiative.
- Eliminated external procurement lead times for a hot press device by reverse engineering a replacement part with GD&T compliant CAD drawings and validating tolerances with machine shop technicians for in-house manufacturing.
- Created a comprehensive repair plan for a malfunctioning injection molding device by independently diagnosing hardware errors and collaborating directly with OEM technical support to retrieve documentation.
- Co-authored a peer reviewed paper on the moisture durability of Western Red Cedar ([DOI: 10.3390/ma18184353](https://doi.org/10.3390/ma18184353)) by analyzing materials testing data and generating technical visualizations using Python libraries.

ENGINEERING DESIGN TEAMS

UBCO Steel Bridge

September 2025 – May 2026

Fabrication Team Lead

- Streamlined the manufacturing and assembly process by translating conceptual structural designs into high-fidelity SolidWorks models, utilizing Weldment and Sheetmetal tools to generate automated cut lists.
- Ensured structural stability and load-bearing integrity for the annual competition by leading the mechanical design of the bridge joints to meet strict specifications.
- Upskilled the fabrication sub-team in generating viable manufacturing drawings by facilitating weekly SolidWorks training sessions and design review meetings.

OK Motorsports FSAE

September 2023 – August 2025

Vehicle Dynamics Engineer

- Optimized the driver interface and vehicle dynamics by collecting and analyzing live performance data during track tests.
- Manufactured custom race car components to precise tolerances by operating in-house machine shop equipment and applying strict metrology techniques.
- Developed a functional pedal box for the driver training simulator by designing the part and creating GD&T compliant manufacturing drawings in Siemens NX.

TECHNICAL PROJECTS

Portable Pneumatic Press | <https://lnkd.in/p/gEwTwQEZ>

2026

- Designed and manufactured a custom safety shield and sliding door housing for a functional pneumatic press using sheet metal, acrylic, and 3D printed components.
- Prevented final assembly bottlenecks by actively coordinating dimensional requirements with other student teams to navigate strict timeline dependencies.
- Documented the complete design lifecycle, 3D printing tolerance analysis, and material selection trials in a comprehensive 90 page technical report adhering to IEEE formatting guidelines.

Ember Alert | <https://youtu.be/prgUgCiNibc>

2024

- Awarded 1st Place Design in 2023-24 APSC 171 Design Competition.
- Engineered an innovative solution to assist firefighters with finding underground embers post wildfires.
- Conceived an all-in-one solution leveraging an ESP32 microcontroller hosting a web app to allow multiple users to access temperature and geolocation data being recorded by an advanced handheld sensor.
- Used SolidWorks to model enclosures, HTML, CSS, and JavaScript used for a custom web app for real-time data visualization.

AE86 SolidWorks Model | <https://youtu.be/9Xse4sq8zzQ>

2024

- Produced a model of the Toyota AE86 with an emphasis on attention to accurate details, totaling to over 300 hours of work in SolidWorks.
- Final assembly incorporated over 20,000 parts including a working drive train, suspension, engine, and transmission.
- Developed comprehensive file management and assembly hierarchy rules to manage 400+ unique parts.

Laptop SSD Recycling

2023

- Developed a novel solution to address waste of aging or dead laptops by repurposing into USB mass storage devices
- Engaged in extensive use of Gantt software to model critical paths and risks and assign tasks ensuring consistency of progress.
- Utilized OpenLCA to conduct a product life cycle analysis.
- Created weekly design reports made in Microsoft word on various stages of design process

CERTIFICATIONS

SOLIDWORKS Design Professional (CSWP)
SOLIDWORKS Design Associate (CSWA)
SOLIDWORKS Additive Manufacturing Associate

Certificate ID: [C-6T2VRD8HP4](#)
Certificate ID: [C-WASR2BPM6Y](#)
Certificate ID: [C-BREF4QHPXA](#)

RECOGNITIONS & AWARDS

Deputy Vice-Chancellor Scholarship for Continuing Students
Dean's List
1st Place Design 2023-24 APSC 171

2024, 2025
2024, 2025
2024

OTHER EXPERIENCE

Pacific National Exhibition

May 2023 - September 2024

Seasonal Attendant

- Managed high-volume logistics for a 3000-car daily parking facility and utilized conflict resolution skills and quick thinking to ensure positive guest experiences.

Science World

Summer 2022

Assistant Science Facilitator

- Managed displays and galleries and helped with opening and closing by ensuring areas were orderly and all items were orderly.
- Strengthened customer service skills by engaging with families and assisting visitors.

PROFESSIONAL ASSOCIATIONS & MEMBERSHIPS

Engineers and Geoscientists British Columbia (EGBC)
SAE international | Collegiate Member (FSAE)

2023 – Present
2023 – 2025